

A certified abstention band for card-fraud decisions, and two quantum arms reported as measured

Global Quantum + AI Challenge 2026 — HSBC track

Code and every table behind this document:

github.com/thedaemon-wizard/Global_Quantum_AI_Challenge_HSBC

1 The decision, and what is missing from it

An issuer authorising a card transaction has three actions, not two: approve, decline, or challenge the cardholder through 3-D Secure. The middle one is what makes the problem tractable at scale — routing even one percent of authorisation volume to human analysts is not operationally possible, while a step-up challenge is. Fraud models are reported by AUC or average precision; neither is a statement about a decision, and neither carries a finite-sample guarantee. Conformal methods reach this problem closely — Singh et al. pair class-conditional prediction with cost-controlled abstention, Zhu et al. certify a false-negative rate on card fraud, Nayak and Bushara report a fraud objective their own paper calls conformal-style rather than a finite-sample guarantee, and Chen et al. carry the temporal case to a graph model — but none conditions the guarantee on the abstention region it routes to, and those that do order time certify a prediction set rather than a decision. Both are what this submission adds, and the second turns out to matter.

What we contribute. A three-valued rule whose *abstention band* carries a distribution-free, finite-sample risk certificate under a pre-registration frozen before any model was fitted, and an honest measurement of where two quantum approaches sit against a tuned classical baseline on the same decision.

What a successful PoC would demonstrate. A step-up decision whose false-decline rate carries a distribution-free, finite-sample certificate at a level the bank sets, validated on a block it never saw — the artefact a model-risk function can audit, which an AUC cannot provide. This study meets that target on public data, at loose levels only ($\alpha \geq 0.10$) and at the band budgets the sample size permits; the PoC target is the same property at the budget the business chooses rather than the one the data forces.

Expected impact. The deliverable is an operating guarantee, not a score, and three consequences follow from that — none of which needs a quantum result. A model-risk function can review a bound instead of a metric, which is what supervisory revalidation SR 26-2 asks of a decisioning model and an AUC cannot supply — and it matters more here, not less, because fraud detection sits in the EU AI Act Annex III *exception*, so no external high-risk regime supplies that discipline. The operating point becomes an input: α and the band budget are chosen by the business, and the certificate reports whether that choice is attainable rather than asserting it. And the band bounds what the expensive component costs — Section 5 prices it against the authorisation budget — which is what makes a quantum re-ranker discussable at all. We attach no monetary figure: issuer-side false-decline costs are not public, so the study certifies the *rate* and leaves the bank to multiply it by a number only the bank has.

Neither quantum arm produced a usable result, and they failed differently. A fidelity quantum kernel was rejected by its own a-priori screens before it was run on the task — a decisive negative. A matrix-product-state classifier ran and did not beat gradient boosting, but that comparison is underpowered, so what it yields is a non-superiority bound rather than a null result. We report both in the body rather than the appendix, because a screen that only ever passes is not a screen, and because the pre-registration committed us to reporting whatever it returned.

What we do not claim. No quantum advantage of any kind. No portfolio-level PSD2 compliance figure — the SCA-RTS ceilings govern exemption eligibility, not a decline threshold, and the appendix records that category error as one we made and withdrew. And no claim that the certificate binds the *unconditional* false-decline rate: it does not, and Section 2 says why that is the point rather than a caveat.

2 Method

Paradigm. Hybrid classical-quantum: a *gate-based* fidelity kernel and a *quantum-inspired* matrix-product-state classifier, both confined to an abstention band and wrapped in a classical certificate that depends on neither.

The rule, and which block may touch which parameter. Figure 1 is the whole method. A classical scorer f splits traffic three ways against thresholds fixed in advance; the band is $B = \{x : \tau_o \leq f(x) < \tau_{hi}\}$ and g is the in-band re-scorer, where a quantum model would enter if one were usable. None was, so the certified rule takes $g = f$ and λ is a third threshold on the same score. Which block sets what carries the guarantee: edges and

thresholds from $\mathcal{D}_{\text{band}}$, λ certified on $\mathcal{D}_{\text{cal}}$, nothing selected on $\mathcal{D}_{\text{test}}$.

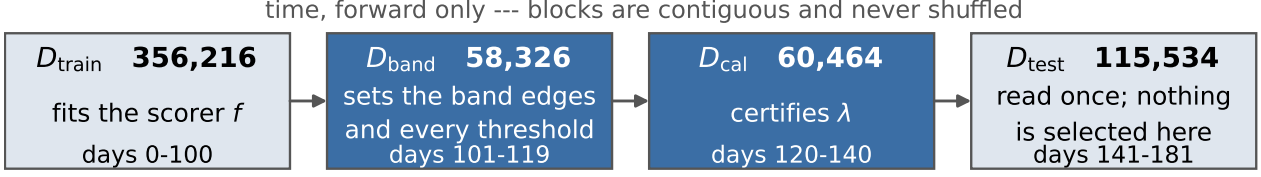


Figure 1: The four-block temporal split; shaded blocks carry the guarantee. Counts and day ranges come from `splits.csv`.

The estimand is band-conditional, deliberately. We certify

$$R(\lambda) = \mathbb{P}(D(X) = \text{DECLINE} \mid Y = 0, X \in B),$$

not the marginal false-decline rate. The marginal rate is dominated by τ_{hi} and would be almost unchanged if g were replaced by a coin flip. A certificate that does not bind the component it is said to license is not evidence about that component — so the harder, smaller-sample quantity is the one we bound, and the cost of that choice is reported in Section 3 rather than hidden.

Two distinct breaks of exchangeability, and the split separates them. The blocks are 60/10/10/20%, contiguous and snapped to day boundaries. The **temporal** break is that $\mathcal{D}_{\text{cal}}$ precedes $\mathcal{D}_{\text{test}}$. The **selective** break is that band membership is a data-dependent event, if the edges were estimated on the data used to certify. Freezing the edges on $\mathcal{D}_{\text{band}}$ removes the second by construction: membership becomes a fixed measurable predicate, and filtering a sequence by one preserves exchangeability of what remains. Only the temporal break survives, and we measure it.

Risk control. Learn-then-Test controls family-wise error over a finite grid fixed in advance, using Hoeffding–Bentkus p -values with Holm. Two risks are controlled jointly: $R(\lambda)$ at level α , and the false-negative rate at α_{FN} .

The bound applies only to a *mean of per-observation losses*. Our first formulation was a nonlinear transform of a mean and did not produce valid p -values; it is now a genuine 0/1 loss averaged over frauds, equivalent to a recall floor at $1 - \alpha_{\text{FN}}$.

Coverage is judged against the exact law, not against α . For a sound split-conformal system the test-error count follows $E \sim \text{BetaBinomial}(m, n+1-k, k)$ with $k = \lceil (n+1)(1 - \alpha) \rceil$. Checking $\hat{r} \leq \alpha$ is a one-sided test against the wrong null: on a correctly calibrated system it passes only about half the time. Every verdict below is a two-sided Beta-Binomial interval.

Intervals resample entities, not rows. The IEEE-CIS label propagates a reported chargeback to later transactions sharing an account, email or billing address, so rows are clustered: 85.0% of test-block `card1` values also appear in training. Every interval is a card-level block bootstrap.

3 Results

What a random split buys you, measured. Before any conformal machinery, the split itself. Table 1 runs one gradient-boosted model at fixed hyperparameters over five seeds on three splits of the same file. A stratified random split reports 0.3164 more average precision than a forward holdout — 0.8254 against 0.5090 — with non-overlapping ranges. That is the largest effect in this study, larger than anything either quantum arm could have contributed, and it is why what follows reports a temporal split.

It is also what separates this baseline from the published leaderboard. The 1st-place IEEE-CIS solution reports ROC AUC 0.9459 on the private leaderboard — the competition’s own held-out file — and about 0.9363 in time-based cross-validation over the training file, against 0.8837–0.8884 here on a strict forward holdout of that training file. Different tasks, and the difference is measured above rather than assumed; the appendix carries the primary metrics in full.

Table 1: The same gradient-boosted model, the same hyperparameters, five seeds, evaluated on the test block of three different splits. One model on one file, so it establishes the size of the effect here rather than a general rule, and it is the control arm of the split design rather than an extra experiment.

Split	ROC AUC	AP	AP vs forward holdout
stratified	0.9658	0.8254	+0.3164
temporal	0.8851	0.5090	—
card-disjoint	0.8509	0.5394	+0.0304

Table 2: Coverage across five seeds per arm and level, judged against the exact Beta-Binomial interval. It replaces a single-split table that reported an across-seed maximum as a centre; the appendix records that retraction. The temporal arm breaches on every seed at the two loosest levels; the stratified arm breaches on none at any level.

Arm	α	Ratio (mean)	sd	Seeds outside
temporal	0.010	1.436	0.055	5 of 5
temporal	0.005	1.456	0.036	5 of 5
temporal	0.002	1.314	0.067	4 of 5
temporal	0.001	0.943	0.020	0 of 5
stratified	0.010	1.008	0.038	0 of 5
stratified	0.001	0.962	0.140	0 of 5
card-disjoint	0.010	0.916	0.133	2 of 5
card-disjoint	0.001	0.882	0.362	1 of 5

Coverage fails under time, on every seed, and the controls do not reproduce it. Table 2 calibrates on $\mathcal{D}_{\text{cal}}$ and reports on $\mathcal{D}_{\text{test}}$, five seeds per arm and level, each judged against the exact Beta-Binomial interval. The temporal arm falls outside on *every* seed at the two loosest levels and on 4 of 5 at the third, and inside on every seed at the tightest; the stratified arm falls outside on none, at any level.

The card-disjoint arm is not clean, and the two levels fail differently. At $\alpha = 0.01$ it is outside on 2 of 5 seeds and both *over*-cover — a two-sided fit failure, not a one-sided breach. At $\alpha = 0.001$ it is outside on 1, and that seed runs the other way: ratio 1.5245 against four below one, the lowest 0.6502. One against four reads as noise, but it is under-coverage, so this arm is not a pass; its spread (sd 0.362) is an order of magnitude wider than the temporal arm’s. Neither control reproduces the temporal arm’s every-seed breach, which is weaker than “the breach is temporal” because the arms differ in more than time. Label censoring is ruled out separately: the trailing-window fraud rate is 3.666 % against 3.281 % earlier — higher, not lower.

But the breach is origin-dependent, and this narrows the claim. Re-running at five rolling calibration origins gives Table 3: the ratio ranges from 0.684 to 1.411. 2 of 5 breach, and the earliest is *conservative* — the deviation reverses direction. So the defensible statement is not that temporal splits inflate the false-decline rate by a fixed factor, but that the deviation depends on where the calibration window falls, and a single origin cannot establish its size.

And it holds on the held-out block. This is H5. All 5 certified configurations were applied to $\mathcal{D}_{\text{test}}$ unchanged — same edges, same λ , no recalibration — and 5 of 5 realised a band-conditional false-decline rate at or below their

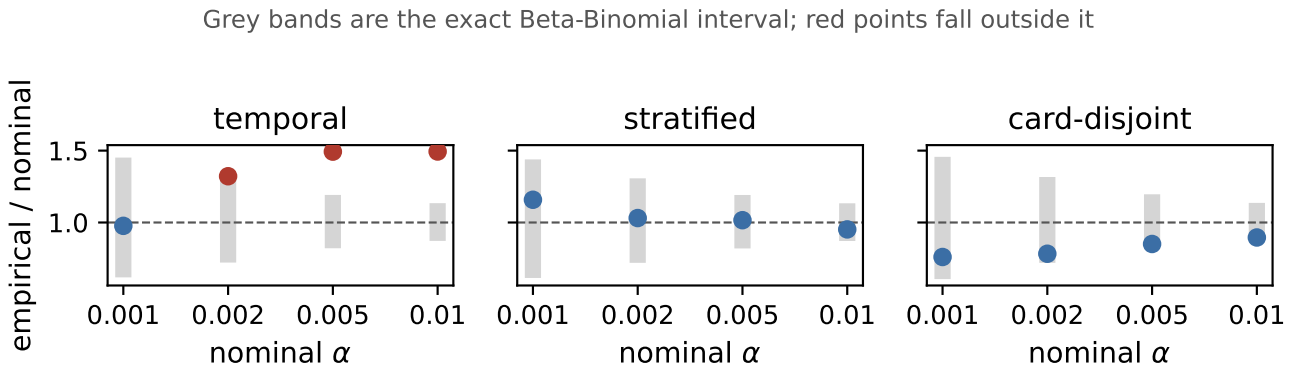


Figure 2: One seed, with the exact Beta-Binomial interval to scale; Table 2 gives all five. The interval widens sharply as α tightens, which is why the temporal arm’s compliance at the tightest level is weak evidence.

Table 3: The temporal deviation across five rolling calibration origins. Two of five breach and the earliest is conservative, so no single ratio characterises the effect.

Origin (day)	Ratio	Errors	Verdict (interval)
101	0.684	379	conservative ([472, 641])
111	0.893	459	inside ([437, 595])
121	1.411	777	breached ([468, 638])
131	1.198	696	breached ([494, 674])
141	1.126	607	inside ([457, 625])

Table 4: H4 in the band: matrix-product-state against the tuned gradient-boosted baseline on identical rows and features. Intervals are 95 % card-clustered bootstraps. Every interval contains zero, and the comparison is underpowered against its pre-registered ceiling, so what is established is a non-superiority bound rather than a null.

χ	AP (MPS)	AP (GBDT)	Δ AP	95 % CI	p
4	0.1202	0.1407	-0.0205	[-0.0593, +0.0149]	0.864
8	0.1173	0.1407	-0.0234	[-0.0630, +0.0122]	0.895
16	0.1183	0.1407	-0.0225	[-0.0613, +0.0136]	0.884
32	0.1235	0.1407	-0.0172	[-0.0582, +0.0200]	0.806

certified α , the tightest using 0.856 of its budget. Holding a ceiling that loose is a real check that the machinery transfers and a weak one, which amendment A1 predicted from the band’s sample size before the arm ran.

What the certificate actually reaches. 5 of 48 configurations certify, all strictly inside the band — the smallest margin below τ_{hi} is 0.0136. That is not automatic: a λ at the band edge flags nothing, so its risk is structurally zero and the guarantee vacuous. An earlier version of this table was exactly that.

The survivors need $\alpha \geq 0.10$ and $\alpha_{FN} = 0.45$, and nothing certifies at the tightest band budget — where $\mathcal{D}_{cal}$ contributes only 847 legitimate in-band rows against 58,343 overall. The binding mechanism is sample size acting through the concentration bound, not the class-conditional degeneracy floor.

4 Two quantum arms, and how each failed

The kernel never reached the task. Before running a fidelity quantum kernel we pre-registered screens over 120 configurations of encoding $\times$ qubits $\times$ bandwidth $\times$ entanglement, and gated on two: a kernel is usable only if it is not exponentially concentrated — effective rank $r_{eff} = \exp(-\sum_i p_i \log p_i)/n$ at $p_i = \sigma_i / \sum_j \sigma_j$, inside a usable band — *and* not reproducible by a tuned RBF, where ρ_{RBF} is the largest absolute off-diagonal correlation over a bandwidth grid and must fall below 0.60.

28 of 120 pass conditioning. **0 pass both.** The closest any configuration came to distinctness was $\rho_{RBF} = 0.6291$; the median is 0.9740 and the maximum 1.0000, indistinguishable to floating point. On 300 stratified rows with at most eight features, no configuration cleared the pre-registered bar — which reporting a downstream accuracy would have hidden. One caveat: the protocol names the Huang geometric difference first and the implementation computes it without gating on it (amendment A6); since Huang treats a *large* difference as favourable, gating would have made this rejection less decisive, not more. The closest published comparator agrees: a controlled benchmark on real card-transaction data (Faryad, arXiv:2608.15718, 2026) reports no robust quantum advantage, and finds one apparent advantage to be an artefact of unequal search budgets.

The tensor network ran, and did not win. A matrix-product-state classifier contracts

$$f_\ell(\mathbf{x}) = \sum_{\{s\}} [A^{s_1} \cdots A^{s_N}]_\alpha W_{\alpha\ell} \prod_j \phi^{s_j}(x_j), \quad \phi(x) = [\cos \frac{\pi x}{2}, \sin \frac{\pi x}{2}],$$

where W is the head that closes the final bond into the two class logits ℓ , sweeping the bond dimension χ rather than tuning it. In the band, on identical rows and features, it reaches average precision 0.1235 at the largest bond dimension against 0.1407 for the tuned gradient-boosted baseline, a paired difference of -0.0172 .

At full scale it loses by a wider margin, with replication behind it. On all 431 features, trained on $\mathcal{D}_{train}$ and scored once on $\mathcal{D}_{test}$, across 16 fits — four bond dimensions at four seeds each — the best draw reaches 0.2512 average precision against 0.5055–0.5114 for the tuned baseline. Every one of the 16 loses by a factor of two or more at a 3.412 % positive rate, so which draw is reported does not change the conclusion.

Capacity is not resolvable, and replication is how we know. The largest spread across seeds at fixed bond dimension is 0.1781 average precision; the spread of the seed *means* across bond dimension is 0.0629 — seed noise is 2.83 times the capacity signal, so a per- χ ordering is selection on a maximum. **And 2 of 16 fits never left chance**, both at one seed and at two different bond dimensions, on the sequential contraction — the stable path. Failure to train at that rate is a property of this ansatz on this data, and a reader choosing a method for a payment system should be told.

The sweep is affordable because cost does not follow the bond dimension the way the ansatz suggests: the jobs took 2747 to 3145 seconds against the 64-fold spread a χ^2 model predicts. A 431-site chain is bound by the launch overhead of its sequential contractions, not their arithmetic.

The comparison is underpowered, and the gate that should have said so was itself wrong. It is a comparison between model *families*, whose measured standard error is 0.0199 — 8.03-fold the baseline-against-itself figure the gate originally used. The effect it can resolve is 0.0557 average precision, against a pre-registered ceiling of 0.023; the appendix gives both defects in full. So H4 is reported as underpowered, which is what the protocol commits to. Every interval contains zero, so any improvement is bounded above by +0.0200 average precision — a non-superiority bound, not evidence of equivalence and not evidence that the tensor network is worse.

5 Feasibility, resources, and the constraints we already know

What the PoC sprint would build. The same protocol on the issuer’s own authorisation stream: fit the incumbent scorer, freeze four blocks, calibrate the band, certify the band-conditional false-decline rate at a level the business picks. Every component has been run end to end here, so the sprint is a port and a re-calibration rather than a build: three stages of roughly a month with a month of slack — split integrity, then calibration and certification, then the quantum re-ranker, last because it is the only stage that can fail to produce a result.

Compute, measured rather than estimated. The classical core is cheap: the tuned baseline fits the in-band problem in 0.28 seconds, and calibration is a sort. The expensive component is the quantum-inspired arm: 16 full-scale tensor-network fits over 356,216 rows and 431 sites took 2747–3145 seconds each on the one RTX PRO 6000 Blackwell (96 GB), CUDA 13, Python 3.12 workstation every number here was measured on. Per-step cost is flat in the bond dimension, so capacity is nearly free, chain length is the cost, and a resource estimate scales with the feature count rather than the ansatz. No QPU is required (Section 7).

Latency, against the budget the statement gives. Of the 300 ms authorisation ceiling about 130 ms is network, leaving roughly 170 ms issuer-side. Scoring one authorisation costs 0.0841 ms, 0.3197 ms at the tail; one in-band kernel evaluation would cost 96.577 ms — 1148 times as much, its tail taking 75.9% of that budget. The classical core is therefore not the latency risk and the kernel is. That kernel figure prices a fixed 64-row support set and is linear in it, so a deployment would have to hold the support set to that order; the band bounds aggregate kernel compute, not per-request latency, which a transaction landing in the band pays in full. Both figures bound model cost on this workstation, single-threaded, rather than production latency.

Data, and why the PoC cannot reuse this file. The method needs settled labels and an entity key to cluster on; both are ordinary in an issuer’s data and only approximated here, and the benchmark’s competition rules permit non-commercial research use only. The protocol ports; the data does not.

Constraints we have measured, and would carry into Phase II. The population is conditional on an incumbent: this file contains only transactions an existing stack approved, because declined attempts cannot generate a chargeback. That is exactly the population a champion/challenger comparison runs on, so it is the right restriction — but no result extrapolates past it. The label is partly an entity flag, since a chargeback propagates to later transactions sharing an account, email or billing address, and Shapley attribution makes that concrete: 0.758 of the in-band model’s total contribution falls on `card1` alone, across 8 features. A governance review would read that as a model which mostly recognises cards — the strongest argument for running this on an issuer’s own data, where an entity key is a key and not a predictor. The certificate is sample-starved by design: conditioning on a narrow band is what makes the guarantee bind the component we care about, and also what leaves 847 legitimate rows to certify with at the tightest budget — the configurations that do certify had 1572 or more. A larger calibration block or a wider band buys tighter α ; the method is unchanged.

What would change the quantum conclusion. An encoding demonstrably outside the RBF family, at a bandwidth that does not concentrate, would pass the same screens unmodified; we did not find one. For the tensor network the requirement is a comparison powered for cross-family noise — about 7.77 times the in-band evaluation set, which a wider band or a longer window supplies. And the negative is about aggregate discrimination, not the operating point: H2, which would have tested decisions at the certified threshold, never ran, and a method that does not improve ranking can still improve one at a low false-positive threshold — the regime this certificate operates in.

6 Validation plan

The plan below is not aspirational: every step was executed in this study on public data, and the results are in Section 3. What a PoC sprint changes is the data, not the protocol. Some pre-registered items were *not* executed — two kernel hypotheses and the entanglement ablation, closed by the screens, and amendment A7’s drift measurements — and the appendix enumerates them, because a plan that cannot say which of its own steps ran is not one.

Success is a certificate, not a score. A PoC succeeds if the band-conditional false-decline rate is certified at a level the business names, with family-wise error control, and the certificate then holds on a block never used to build it. Average precision and ROC AUC are reported alongside but decide nothing. Failure is equally well defined — the admissible set comes back empty, which is what happens here at the tightest band budget — and the acceptance test is the held-out read reported in Section 3.

Size the evaluation block to the power target first. A detectable-effect ceiling is fixed *before* the comparison and the block sized to it, which is the one change that would make a quantum-versus-classical comparison here informative; Section 4 reports what happens when it is not.

Threshold change management. λ moves on every recalibration, so it needs an owner and a cadence rather than a rerun. Model risk approves each move quarterly, matching the rolling window the fraud-rate definition already uses, and out of cycle whenever realised in-band risk crosses the certified α . If a recalibration fails to certify, the rule falls back to the previously certified λ , and failing that to the incumbent scorer with the band closed — degrading to today’s behaviour rather than to an uncertified threshold. Phase II runs the challenger in shadow first, scoring every authorisation and acting on none.

What keeps the plan honest. Grids, splits and thresholds are hashed before any model is fitted, the hypotheses are frozen with dated amendments, and every retraction is recorded — 89 decision entries, including ones that removed results already written down. Reproduction is `make smoke && make reproduce && make check`.

7 Where the quantum method sits, and why there

Classical, on all traffic. A gradient-boosted scorer runs on every transaction — the only component that sees full volume, and deliberately the strongest available baseline at 0.8837–0.8884 ROC AUC over five seeds. A hybrid whose classical half is weak proves nothing about its quantum half.

Conformal, on the decision. The certificate wraps whatever scores the band, so the deliverable survives a negative quantum result — as it had to here.

Quantum, on the band only. At a 3.412% positive rate and a 3-D Secure challenge budget the band holds thousands of rows rather than hundreds of thousands, and a kernel whose Gram matrix costs $O(n^2)$ circuit evaluations is affordable on thousands. The band is not a convenience; it is the only place in this pipeline where the arithmetic works.

The compute path. Both arms run on simulators, which the statement permits: no number here comes from a QPU. The fidelity kernel’s reference implementation is an exact statevector, with Braket’s `LocalSimulator` and Aer on CPU and GPU agreeing to within 3.442e-15 over 12 comparisons; the tensor network runs in PyTorch on one GPU. On the sample count the statement asks for: the screens ran on 300 stratified rows per configuration across 120 configurations, and the kernel arm was rejected before any downstream evaluation, so no larger quantum execution took place. The circuits are within near-term reach — at most 8 qubits, transpiled depth 35 and 28 two-qubit gates on a portable basis — so what stopped this arm was the screens, not the hardware.

8 Team

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The statistical machinery is not new to the author. In QIntern 2026 Project 12 (QWorld), within Team A — which owns the paper’s conformal-coverage proposition — he wrote the split-conformal calibration of the false-alarm rate and the results freeze, whose manifests re-verify clean and whose suite was green at 135 on handoff. *Protocol-final and numbers-provisional in the team’s own words — the quantum side runs on a placeholder interface — so it establishes machinery, which is what this submission reuses.*

Prior results. *Tensor networks:* Yale Peaked Hackathon 2026, 13th of 549 at 450 of 550, on a matrix-product-state attack on circuits to 69 qubits. *Quantum kernels:* QPoland 2025 runner-up. *Upstream:* unitaryHACK 2026, merged into CUDA-Q and QuEST.